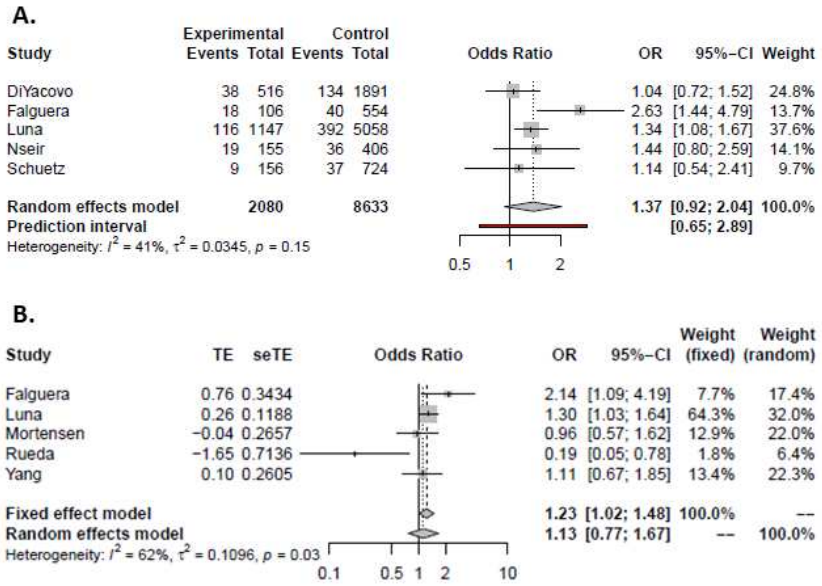
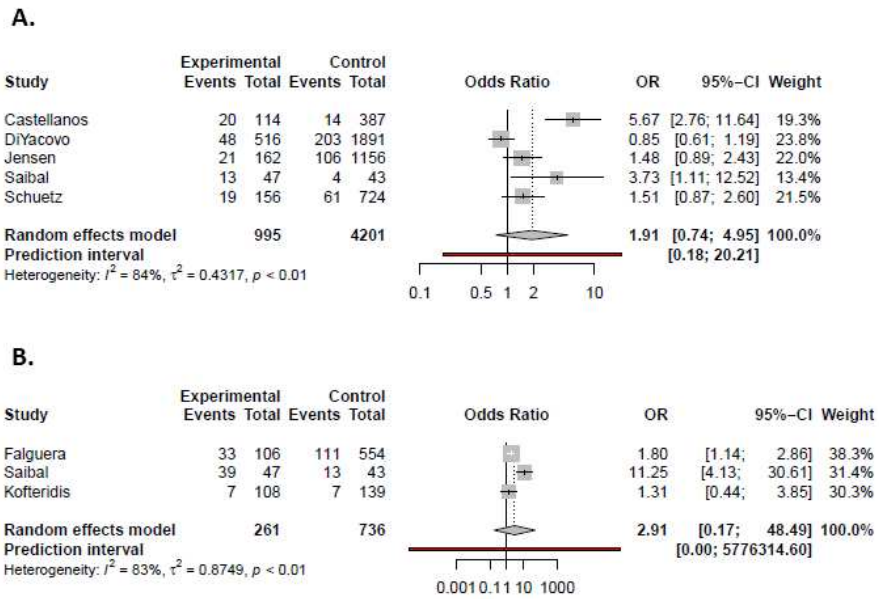
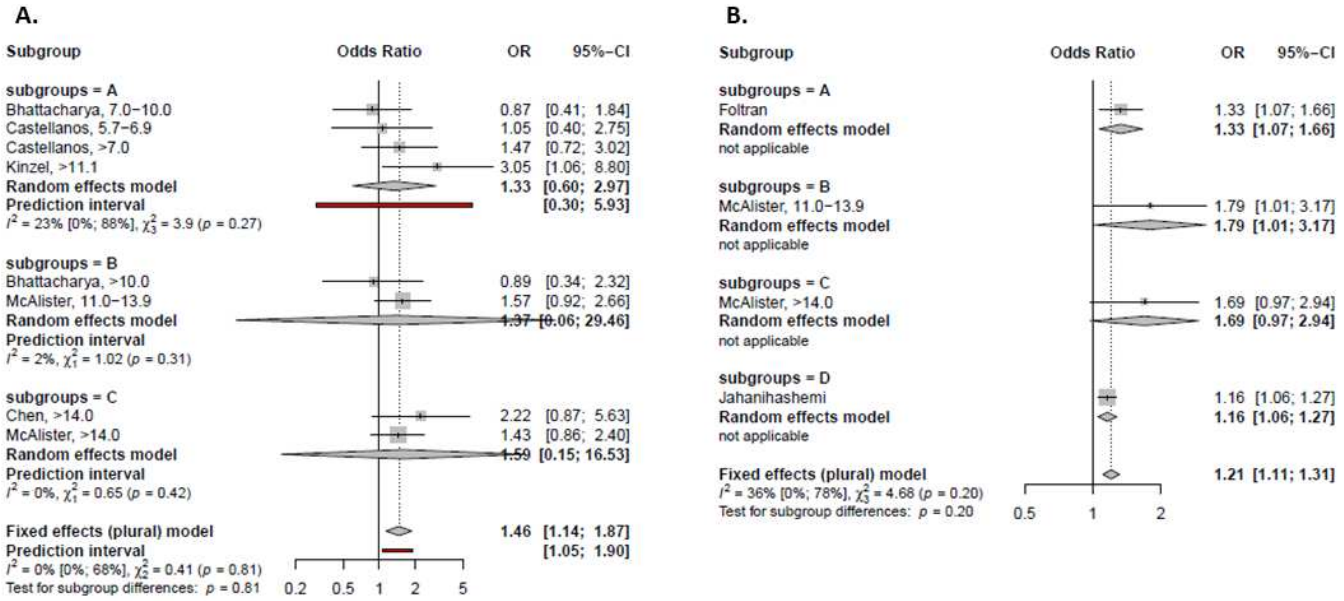
Supplementary Figures



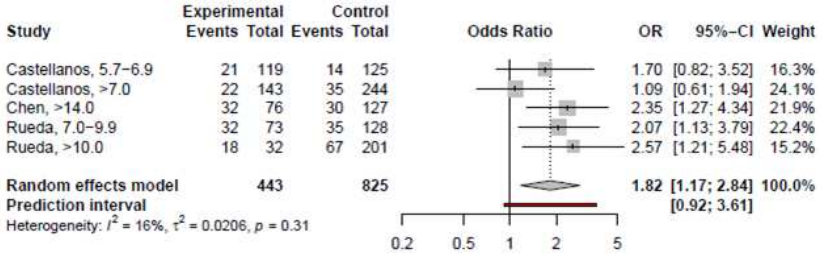
Supplementary figure S1: Meta-analyses of the association between diabetes and 30-day mortality in hospitalized patients with community-acquired pneumonia. Pooled crude (**A**) and adjusted (**B**) odds ratios.



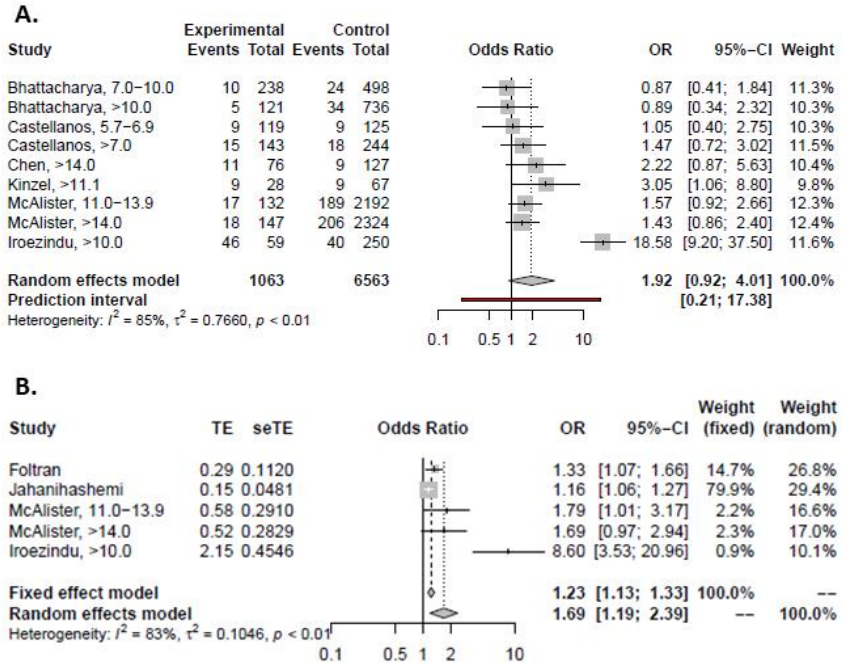
Supplementary figure S2: Meta-analyses of the association between diabetes and intensive care unit (ICU) admission (**A**) and pleural effusion (**B**) in hospitalized patients with community-acquired pneumonia, reported as crude odds ratios.



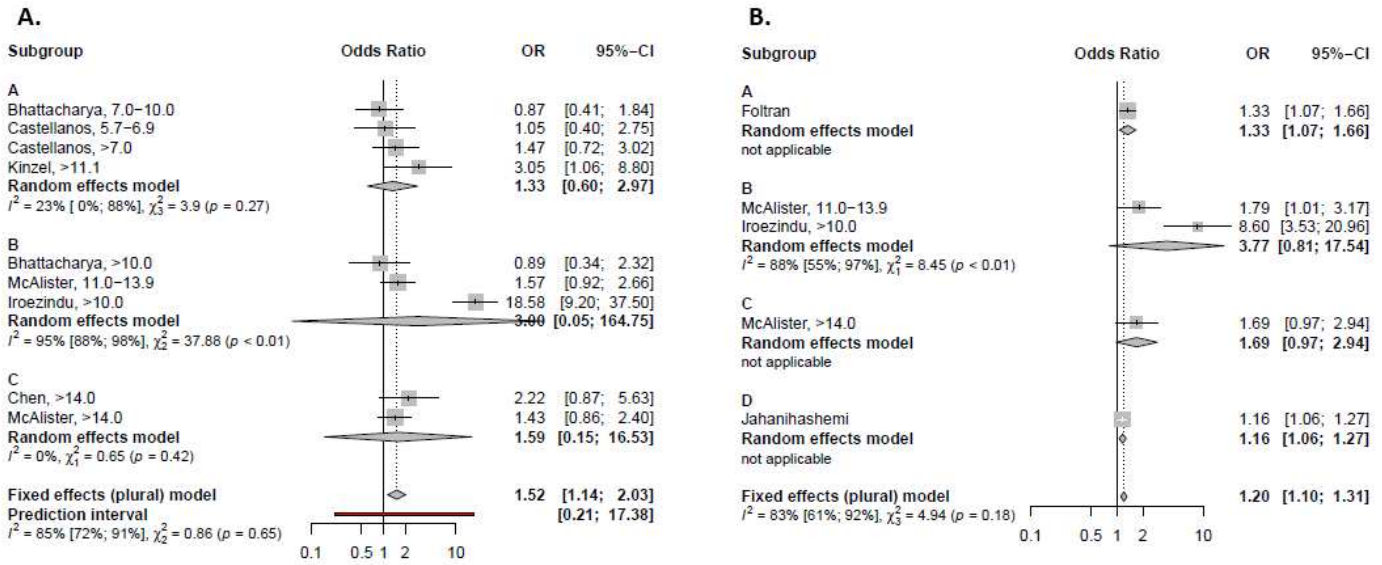
Supplementary figure S3: Meta-analyses of the association between hyperglycaemia and in-hospital mortality in hospitalized patients with community-acquired pneumonia stratified by subgroup. Pooled crude (**A**) and adjusted (**B**) odds ratios by glycaemic cut-off category using the mixed-effects model. Hyperglycaemia subgroup cut-offs = A: ≤ 7.8 mmol/L, B: 7.9-11.0 mmol/L, C: ≥ 11.1 mmol/L, D: not defined.



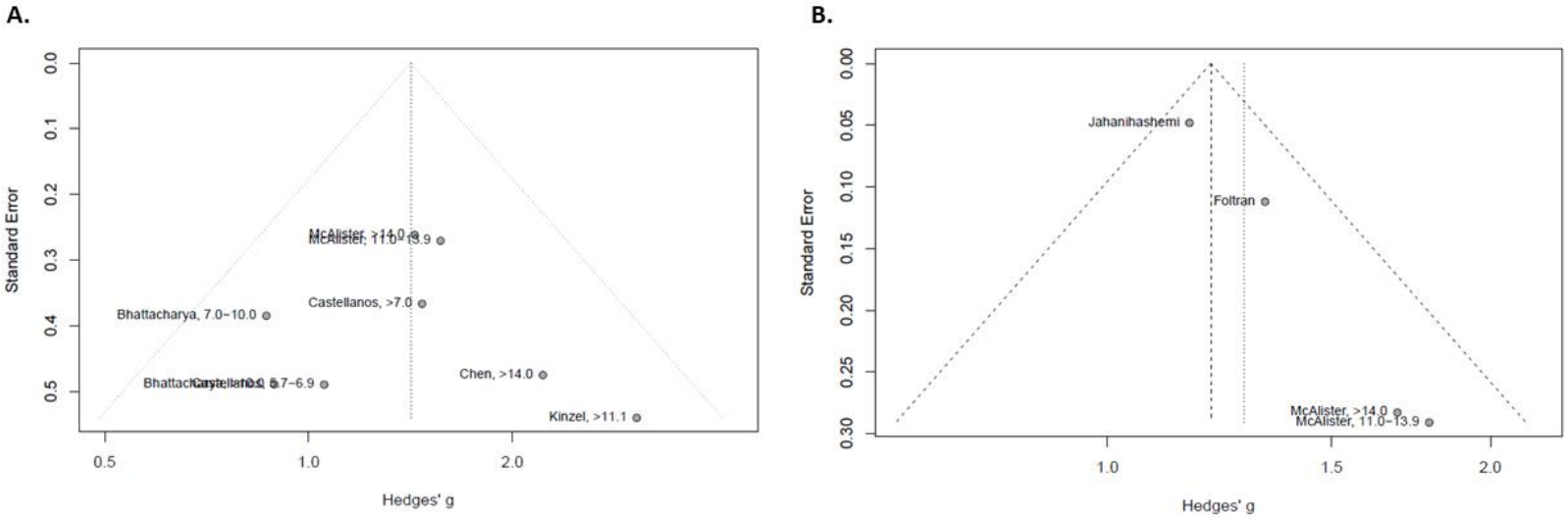
Supplementary figure S4: Meta-analysis of the association between hyperglycaemia and intensive care unit (ICU) admission in hospitalized patients with community-acquired pneumonia for studies reporting crude odds ratios.



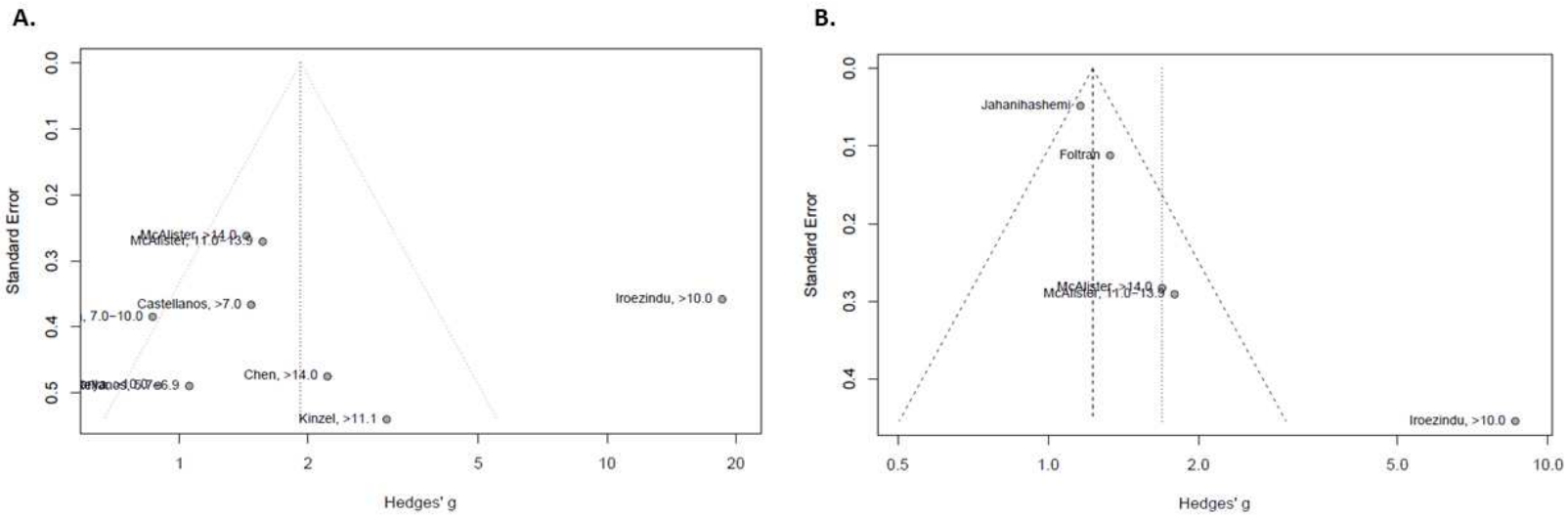
Supplementary figure S5: Meta-analyses of the association between hyperglycaemia and in-hospital mortality in hospitalized patients with community-acquired pneumonia including the study with outlier results by Iroezindu et.al. (24). Pooled crude (**A**) and adjusted (**B**) odds ratios.



Supplementary figure S6: Meta-analyses of the association between hyperglycaemia and in-hospital mortality in hospitalized patients with community-acquired pneumonia stratified by subgroup including the study with outlier results by Iroezindu et.al. (24). Pooled crude (**A**) and adjusted (**B**) odds ratios by glycaemic cut-off category using the mixed-effects model. Hyperglycaemia subgroup cut-offs = A: ≤ 7.8 mmol/L, B: 7.9-11.0 mmol/L, C: ≥ 11.1 mmol/L, D: not defined.



Supplementary figure S7: Funnel plots of the association between hyperglycaemia and in-hospital mortality in hospitalized patients with community-acquired pneumonia for studies reporting crude (A) and adjusted (B) odds ratios.



Supplementary figure S8: Funnel plots of the association between hyperglycaemia and in-hospital mortality in hospitalized patients with community-acquired pneumonia including the study with outlier results by Iroezindu et.al. (24) for studies reporting crude (**A**) and adjusted (**B**) odds ratios.

Supplementary Tables

First author, year, country	Funding source	Study design	Quality (NOS Score)	Data source	Primary outcome	Inclusion Criteria	Exclusion Criteria	Length (yrs), period
Adamuz, 2014, Spain(19)	Fondo de Investigacion Sanitaria de la Seguridad Social and others	Cohort: Prospective	8: High	Hospital records and the SAP Healthcare Database of the Catalan Health Service	1-year mortality after hospital discharge	Adult patients >17 years of age with CAP admitted to hospital	Immunosuppression, in-hospital mortality.	5.0, 2007-2011
Benfield, 2007, Denmark(52)	Danish Heart Foundation; Danish Medical Association Research Fund and others	Cohort: Retrospective	9: High	Copenhagen City Heart Study, Danish National Hospital Discharge Register, Danish Civil Registration System National Register of Causes of Death	Risk of infectious disease hospitalization; secondary outcome of mortality after hospitalization for given infectious disease (e.g. pneumonia)	Patients >16 years of age with a diagnosis of pneumonia admitted to hospital	Unclear	11.0, 1991-2001
Bhattacharya, 2013, United States of America(38)	None reported	Cohort: Retrospective	9: High	Hospital records	In-hospital mortality, length of stay, and readmission for community-acquired pneumonia within 30 days	Age ≥ 40 years with CAP admitted to hospital	Missing glucose measures, disposition code, or race data.	3.0, 2008-2010
Biteker, 2016, Turkey(53)	None reported	Cohort: Prospective	6: Moderate	Prospective collected data on admission to ED	Complicated hospitalization (CH) defined as intensive care unit admission, need for mechanical ventilation, or in-hospital mortality	Age ≥ 18 years with CAP admitted to hospital	Age < 18 years, active pulmonary TB, hospital-acquired pneumonia, pregnancy, immunosuppression, chronic dialysis.	0.3, 2015-2015
Biteker, 2018, Turkey(54)	None reported	Cohort: Prospective	9: High	Prospectively collected	Complicated hospitalization (CH) defined as intensive care unit admission, need for mechanical ventilation, or in-hospital mortality	Age ≥ 18 years with CAP admitted to hospital	Age < 18 years, active pulmonary TB, hospital-acquired pneumonia, pregnancy, immunosuppression, chronic dialysis.	1.0, 2015-2016
Cangemi, 2015, Italy(22)	Grant from the Sapienza University of Rome	Cohort: Prospective	7: High	Prospectively collected; Follow-up data from review of hospital databases, medical records, death certificates, or telephone interviews	Death from any cause at long-term follow-up	Age ≥ 18 years with CAP admitted to hospital	Pre-existing radiologic infiltrates, immunosuppression, malignancy, pregnancy or breastfeeding, health care-associated pneumonia, antibiotic allergy, no consent.	4.0, 2011-2014
Castellanos, 2010, United	None reported	Cohort: Retro-	8: High	Hospital records	Pneumonia complications, defined as “Mild” or “Severe”	Age ≥ 18 years with CAP admitted to hospital and	Hospital-acquired pneumonia, admission to critical care unit.	1.0, Unclear

States of America(39)		spective				early morning fasting blood glucose level drawn within 24 hours of admission		
Chen, 2015, Taiwan(35)	None reported	Cohort: Retro-spective	7: High	Hospital records	Adverse outcomes including mortality during admission, ARDS, bilateral pulmonary infiltrates on CXR consistent with oedema, and others	Patients with diabetes and CAP with plasma glucose levels at presentation and HbA1c within 1 month of admission (before or after)	Concurrent infections, steroid use, hypoglycaemia (<70mg/dL), anaemia requiring blood transfusion.	5.3, 2007-2012
DiYacovo, 2013, Spain(28)	Ministerio de Ciencia e Innovacion and others	Cohort: Pros-pective	9: High	Prospectively collected using computer-assisted protocol	Pneumonia outcomes included mortality, length of stay and in-hospital complications, all assessed separately	Age ≥ 18 years with CAP admitted to hospital	Immunosuppression, corticosteroid therapy (20 mg prednisone/day or equivalent) at admission.	8.6, 2002-2010
Eurich, 2010, Canada(33)	None reported	Cohort: Pros-pective	9: High	Prospectively collected using standardized abstraction forms, linkage to provincial administrative databases	Death from any cause within 90 days following hospital admission	Age ≥ 18 years with CAP admitted to hospital, casual blood glucose measurement on admission	Diabetes, admission glucose < 4mM or > 20mM, ICU admission, aspiration pneumonitis, TB, cystic fibrosis, pregnancy, breastfeeding, immunosuppression.	3.0, 2000-2002
Falguera, 2005, Spain(29)	None reported	Cohort: Pros-pective	8: High	Prospectively collected, hospital records	30-day mortality and pleural effusion	Age ≥ 18 years with CAP admitted to hospital	Misdiagnosis, nosocomial pneumonia, no consent, TB, opportunistic infections.	5.0, 1998-2002
Foltran, 2013, Italy(55)	None reported	Cohort: Retro-spective	9: High	Hospital administrative data, laboratory measurement dataset	In-hospital mortality	Patients with CAP admitted to hospital and glucose recorded on admission	Lack of admission glucose record.	3.0, 2003-2006
Godar, 2011, United States of America(36)	None reported	Cohort: Retro-spective	9: High	Hospital records	LOS, 30-day hospital readmission, ICU transfer, 365-day mortality	Age > 17 years at diagnosis, radiologic evidence and symptoms of CAP, at least 1 fasting glucose measure within 6 hours of presentation, admission to hospital	ICU admission, transfer from another hospital, left hospital against medical advice, hospital discharge <30 days pre-admission, pneumonia diagnosed ≥ 48 hours after admission, history of or suspected: aspiration, fungal, or viral pneumonia, pulmonary TB or pneumocystis carinii pneumonia, immunosuppression, tracheostomy.	15.5, 1992-2007
Hifumi, 2015, japan(56)	None reported	Cohort: Retro-spective	9: High	Hospital records	Percentage weaned successfully from mechanical ventilation	Age ≥ 65 years, received mechanical ventilation for CAP in the ED	Recent ICU admission, nursing home or long-term care resident, recent intravenous antibiotics,	7.0, 2006-2012

							chemotherapy, or wound care, attended a hospital or haemodialysis clinic.	
Hraiech, 2013, France(57)	None reported	Cohort: Pros-pective	9: High	Previous prospective cohort study	In-hospital mortality	Age ≥ 18 years with CAP admitted to medical ICU requiring invasive mechanical ventilation	No invasive ventilation, suspected aspiration pneumonitis, TB, or cystic fibrosis, intubation for ≥12 hours, referred from another hospital.	5.0, 2007-2011
Jahanihashemi, 2018, Iran(58)	None reported	Cohort: Pros-pective	9: High	Prospectively performed interview with patients and their companions	In-hospital mortality	Age ≥ 12 years with CAP admitted to hospital	Lung cancer, history of TB or HIV, admitted to hospital within last month, structural respiratory tract damage.	1.0, 2014-2015
Jensen, 2017, Denmark(34)	The Christenson–Cesons Family Foundation and others	Cohort: Retro-spective	9: High	CAP-NORTH Cohort, Hospital records	Severe outcome, defined as: ICU admittance, in-hospital mortality, or combined risks (ICU admittance and/or in-hospital mortality)	Age ≥ 18 years with CAP admitted to hospital	Recent pneumonia admission, active TB, missing data, immunosuppression.	1.5, 2011-2012
Kabeya, 2016, Japan(27)	None reported	Cohort: Retro-spective	9: High	Inpatient administrative claims database	Length of stay, hospital costs	Age ≥ 65 years with CAP admitted to hospital	Admissions subsequent to the initial admission.	4.0, 2010-2013
Kao, 2017, Taiwan(59)	None reported	Cohort: Retro-spective	9: High	Hospital records	All-cause in-hospital mortality	Admitted for community-acquired Legionnaires' disease and pneumonia	Unclear	4.8, 2012-2017
Kinzel, 1988, United States of America(37)	None reported	Cohort: Retro-spective	8: High	Hospital Records	Mortality and length of stay	Age ≥ 60 years with CAP admitted to hospital	Patients not fulfilling criteria for pneumonia.	2.0, 1982-1984
Koskela, 2014, Finland(21)	Hospital District of Northern Savo	Cohort: Pros-pective	8: High	Prospectively collected	Mortality from 30 days after pneumonia to the end of follow-up	Age ≥ 18 years with CAP admitted to hospital	ICU admission, no consent, antibiotic treatment commenced elsewhere.	1.6, 2006-2008
Lepper, 2012,Germany (18)	German Ministry of Education and Research	Cohort: Pros-pective	8: High	Prospectively collected data from CAPNETZ database	Mortality (28, 90 and 180 days)	Age ≥ 18 years with CAP admitted to hospital	Recent hospital admission, immunosuppression, and active TB.	7.5, 2002-2009
Lopez-de-Andres, 2017, Spain(60)	Fondo de Investigaciones Sanitarias- Health Research Fund	Cohort: Retro-spective	9: High	Spanish National Hospital Discharge Database (Conjunto Mínimo Básico de Datos (CMBD))	Readmission, in-hospital mortality, length of hospital stay	Age ≥ 40 years with CAP admitted to hospital	Type 1 diabetes mellitus	10.0, 2004-2013
Luna, 2016, 17 countries(61)	None reported	Cohort: Retro-spective	9: High	Community-Acquired Pneumonia Organization (CAPO) study database	30-day hospital mortality	Age ≥ 18 years with CAP admitted to hospital	Residence in a nursing home, HIV infection, suspected influenza A H1N1 2009 infection.	10.3, 2001-2011
Martins, 2016, Portugal(23)	Pfizer Grant	Cohort: Retro-	8: High	National Hospital Discharge Database	Mortality and length of stay	Age 20-79 years with CAP admitted to hospital	Recent admission, day cases, hospitalized > 90 days,	4.0, 2009-

		spective					immunosuppression.	2012
McAlister, 2005, Canada(40)	Alberta Heritage Foundation for Medical Research and others	Cohort: Pros-pective	9: High	Prospectively collected	Mortality, non-metabolic complications, cardiac complications, nosocomial infections	Age ≥ 18 years with CAP admitted to hospital	No glucose at presentation, diagnosed or suspected aspiration pneumonia, TB, cystic fibrosis, immunosuppression, pregnant or breastfeeding, ICU admission.	2.0-2000-2002
Mortensen, 2010, United States of America(62)	Department of Veteran Affairs Veterans Integrated Service grant and others	Cohort: Retro-spective	8: High	Hospital records	30-day mortality	Age ≥ 18 years with CAP admitted to hospital	Recent admission, transfer from another hospital, palliative status, subsequent hospitalizations.	4.0, 1999-2002
Nseir, 2019, Israel(63)	None reported	Cohort: Retro-spective	8: High	Hospital records	30-day all-cause mortality	Age ≥ 18 years admitted with CAP and underwent an abdominal ultrasound examination	Hospital-acquired pneumonia, structural lung disease, asthma, empyema, stroke, aspiration pneumonia, pneumonia severity index class I and V, cirrhosis or significant hepatic disease, significant renal disease, alcohol intake > 20 g/day, immunosuppression, malignancy, intravenous drug use, active autoimmune disease, pregnancy.	4.2, 2013-2017
O'Meara, 2005, United States of America(64)	National Heart, Lung, and Blood Institute	Cohort: Pros-pective	9: High	Cardiovascular Health Study (CHS) records, follow-up clinics annually, phone calls, HCFA Medicare utilisation files	Mortality (in-hospital, total mortality after hospitalization)	Age ≥ 65 years with CAP admitted to hospital	Institutionalized, not ambulatory at home, in hospice care, cancer therapy, likely to move in 3 years, no consent, concurrent cardiovascular events.	12.5, 1989-2001
Parrott, 2017, Japan(65)	None reported	Cohort: Retro-spective	9: High	Hospital records	Pathogens, risk factors, and disease outcomes	Age ≥ 18 years with CAP admitted to hospital	Unclear	3.0, 2011-2013
Rueda, 2010, United States of America(31)	Department of Veterans Affairs Merit Review Program and others	Cohort: Pros-pective (Case Control element exclude d)	8: High	Prospectively collected	ICU admission within 72h, mortality at 7 and 30 days	Age ≥ 18 years with CAP due to pneumococcal pneumonia admitted to the hospital	Unclear	7.5, 2000-2007

Saibal, 2012, Bangladesh(24)	None reported	Cohort: Prospective	6: Moderate	Prospectively collected	Pathogens and disease outcomes	Age ≥ 18 years with CAP admitted to hospital	Acid fast bacilli in sputum, bronchial carcinoma, chronic renal disease stage 4 or 5, heart failure, pregnancy.	0.8, 2009-2009
Schuetz, 2014, Switzerland(32)	Swiss National Research Foundation and others	Cohort: Prospective	8: High	ProHOSP study records	Inflammatory response and adverse clinical course (30 day all-cause mortality, ICU)	Age ≥ 18 years with CAP admitted to hospital and blood glucose level taken on admission	Included during earlier hospitalizations, not fluent in German, no consent, immunosuppression, concomitant infection, drug abuse, hospital-acquired pneumonia, palliative status.	1.5, 2006-2008
Suter-Widmer, 2012, Switzerland(25)	Swiss National Science Foundation and others	Cohort: Prospective	9: High	ProHOSP study records	Length of stay	Age ≥ 18 years with CAP admitted to hospital, survived to discharge	Active intravenous drug use, immunosuppression other than corticosteroid use, palliative status, hospital acquired pneumonia, chronic infection needing antibiotic treatment.	1.5, 2006-2008
Yang, 2018, Taiwan(66)	Ministry of Science and Technology, Taiwan and others	Cohort: Retrospective	8: High	Hospital records	30-day mortality	Age ≥ 18 years with CAP and concomitant mono-microbial bacteraemia admitted to hospital	Contaminated blood cultures, incomplete records, hospital-onset bacteraemia, bacteraemia diagnosed prior to ED visit, poly-microbial bacteraemia, uncertain outcome 30 day after bacteraemia onset.	6.0, 2008-2013
Yende, 2010, United States of America(20)	GenIMS funded by NIGMS R01 GM61992 and others.	Cohort: Prospective	7: High	GenIMS study records	All-cause mortality within the first year	Age > 18 years with CAP admitted to hospital	Transfer from another hospital, discharge from hospital in past 10 days, diagnosis of pneumonia in past 30 days, mechanical ventilation, cystic fibrosis, active pulmonary TB, palliative status, prior enrolment, incarceration, or pregnancy.	3.0, 2001-2003
Iroezindu, 2016, Nigeria(30)	Pan-African Thoracic Society Methods in Epidemiologic and others	Case Control	8: High	Hospital records	Predictors of mortality and length of stay	Age ≥ 18 years with CAP admitted to hospital	Hospital-acquired pneumonia, no chest x-ray, findings suggestive of TB, malignancy, or pulmonary oedema.	5.0, 2008-2012
Kofteridis, 2016,	None reported	Case Control	6: Moderate	Hospital records	Mortality and length of stay	Age ≥ 65 years with CAP admitted to hospital	Immunosuppression, malignancy, hospitalized for	7.0, 2005-

Greece(26)							≥ 2 days during the previous 90 days.	2011

Supplementary table 1: General characteristics of the thirty-eight studies identified in the review. NOS = Newcastle-Ottawa Scale. CAP = community-acquired pneumonia. ED = emergency department. ICU = intensive care unit. TB = tuberculosis. HIV = human immunodeficiency virus.

First author, year	Diabetes definition	Numbers			Mortality measure	Other outcomes
		Total	DM	No DM		
Adamuz, 2014	Unclear	1,284	325	959	1-year	N/A
Benfield, 2007	Answering 'yes' to the question: 'Do you have diabetes?'	586	Unc.	Unc.	30-day	N/A
Biteker, 2016	Unclear	111	22	89	N/A	Complicated hospitalization
Biteker, 2018	Unclear	154	33	121	N/A	Complicated hospitalization
Cangemi, 2015	As per European guidelines on cardiovascular disease prevention in clinical practice (version 2012)	301	81	220	End-of-trial	CV events in follow-up
Castellanos, 2010	Unclear	501	114	387	In-hospital	Complicated hospitalization, ICU admission, length of stay
DiYacovo, 2013	Fasting glucose ≥ 7.0 mmol/L (126 mg/dL) on 2 or more occasions, random glucose ≥ 11.1 mmol/L (200 mg/dL), previous clinical and/or biochemical diagnosis, or treatment with oral antidiabetic agents or insulin.	2,407	516	1,891	In-hospital, 30-day	ICU admission, length of stay
Falguera, 2005	Previous clinical and/or biochemical diagnosis, or treatment with oral antidiabetic agents or insulin, fasting glucose ≥ 126 mg/dL (7.0 mmol/L), or random glucose ≥ 200 mg/dL (11.1 mmol/L) on 2 or more occasions.	660	106	554	30-day	Pleural effusion

Hifumi, 2015	Unclear	71	13	58	In-hospital	Weaning from mechanical ventilation
Hraiech, 2013	Unclear	100	18	82	In-hospital	N/A
Jahanihashemi, 2018	ADA criteria	621	68	553	In-hospital	N/A
Jensen, 2017	Self-reported diabetes, attending physician diagnosis, or treatment with either oral or injectable anti-diabetes medication.	1,318	162	1,156	In-hospital	Severe outcome, ICU admission
Kabeya, 2016	Diabetes on discharge coding or registered in the medical record	828	250	578	In-hospital	Length of stay, hospital costs
Kao, 2017	Unclear	32	11	21	In-hospital	N/A
Koskela, 2014	Doctor's diagnosis of diabetes before the current pneumonia episode and verified from the patient files	153	22	131	End-of-trial	N/A
Lepper, 2012	Self-report or drug use, including long term use of glucose lowering drugs	6,891	1,114	5,756	30-day, 90-day, End-of-trial	N/A
Lopez-de-Andres, 2017	ICD-9-CM codes: 250.x0 and 250.x2	901,136	223,715	677,421	In-hospital	N/A
Luna, 2016	Unclear	6,205	1,147	5,058	30-day	N/A
Martins, 2016	ICD-9-CM 250 on discharge coding	74,175	19,212	54,963	In-hospital	Length of stay

McAlister, 2005	Unclear	2,471	401	2,070	In-hospital	Complicated hospitalization
Mortensen, 2010	Unclear	787	230	557	30-day	N/A
Nseir, 2019	Unclear	561	155	406	30-day	N/A
O'Meara, 2005	Fasting glucose of at least 126 mg/dL or the use of insulin or oral hypoglycaemic medications	442	64	378	In-hospital, End-of-trial	N/A
Parrott, 2017	Unclear	97	9	88	N/A	Length of stay
Rueda, 2010	Diagnosis before admission, during admission, or in the 1 year after admission, based on ADA criteria.	233	53	180	30-day	N/A
Saibal, 2012	Current or previous biochemical diagnosis according to WHO definition	90	47	43	In-hospital	Pleural effusion, ICU admission, length of stay
Schuetz, 2014	Self-report or medical record documentation	880	156	724	30-day	Severe outcome, ICU admission
Suter-Widmer, 2012	Self-report or medical record documentation	875	107	768	N/A	Length of stay
Yang, 2018	Unclear	278	93	185	30-day	N/A
Yende, 2010	Self-report and review of medications	1,895	384	1,511	1-year	Severe outcome
Kofteridis, 2016	ADA criteria	247	108	139	In-hospital	Complicated hospitalization, pleural effusion, length of stay

Supplementary table 2: Details of the thirty-one studies associating diabetes status with hospitalized community-acquired pneumonia outcomes. DM = diabetes mellitus. Unc. = unclear. ADA = American Diabetes Association. WHO = World Health Organisation. CV = cardiovascular. ICU = intensive care unit.

First author, year	Hyperglycaemia definition	Numbers			Mortality measure	Other outcomes
		Total	HG	No HG		
Bhattacharya, 2013	Admission glucose: normal glycemia < 126 mg/dL (< 7.0 mmol/L), mild hyperglycemia 126–180 mg/dL (7.0-10.0 mmol/L) [A], and uncontrolled hyperglycemia ≥ 180 mg/dL (≥ 10 mmol/L) [B]	857	A: 238 B: 121	498	In-hospital	Length of stay
Castellanos, 2010	Admission glucose: normal glycemia ≤ 100 mg/dL (≤ 5.6 mmol/L), mild hyperglycemia 101–125 mg/dL (5.7-6.9 mmol/L) [A], severe hyperglycemia > 126 mg/dL (> 7.0 mmol/L) [B], and known diabetes	387	A: 119 B: 143	125	In-hospital	Severe outcome, ICU admission, complications, length of stay
Chen, 2015	Admission glucose ≥ 252 mg/dL (≥ 14.0 mmol/L)	203	76	127	In-hospital	ICU admission, complications, length of stay
DiYacovo, 2013	Admission glucose > 198 mg/dL (> 11.0 mmol/L) [A] or > 252 mg/dL (> 14 mmol/L) [B], only assessed within the cohort with diabetes	516	A: 228 B: 141	288	30-day	N/A
Eurich, 2010	Admission glucose: 72-109 mg/dL (4.0-6.0 mmol/l); 110-139 mg/dL (6.1-7.7 mmol/l); 140-199 mg/dL (7.8-11.0 mmol/l) [A]; and 200-360 mg/dL (11.1 to 20.0 mmol/l) [B]	2,366	A: 535 B: 129	778	90-day, 1-year	N/A
Foltran, 2013	Continuous variable, each 10 mg/dL (0.6 mmol/L) below and above 86 mg/dL (4.8 mmol/L)	1,018	Unc.	Unc.	In-hospital	N/A

Godar, 2011	Admission glucose ≥ 140 mg/dL (≥ 7.8 mmol/L)	969	370	599	1-year	Length of stay
Jahanihashemi, 2018	Not defined	621	Unc.	Unc.	In-hospital	N/A
Jensen, 2017	Admission glucose ≥ 200 mg/dL (≥ 11.1 mmol/L)	1,318	97	1,221	In-hospital	Severe outcome, ICU admission
Kinzel, 1988	Fasting BGL > 140 mg/dL (> 7.8 mmol/L) on at least two occasions or random BGL > 200 mg/dL (> 11.1 mmol/L)	95	28	67	In-hospital	Length of stay
Koskela, 2014	Hyperglycaemia in first 24 hours of hospitalization without a prior diagnosis of diabetes. Fasting hyperglycaemia defined as glucose > 127 mg/dL (> 7.05 mmol/L) between 0300-0700 pre-breakfast. Postprandial hyperglycaemia defined as glucose > 194 mg/dL (> 10.75 mmol/L) during the day or in the evening	131	43	88	End-of-trial	N/A
Lepper, 2012	Admission glucose: < 72 mg/dL (< 4.0 mmol/L), 72-107 mg/dL (4.0-5.9 mmol/L), 108-197 mg/dL (6.0-10.9 mmol/L) [A], 198-251 mg/dL (11.0-13.9 mmol/L) [B], ≥ 252 mg/dL (≥ 14 mmol/L) [C]	6,016	A: 2,768 B: 303 C: 217	2,728	90-day	N/A
McAlister, 2005	Admission glucose: ≤ 109 mg/dL (≤ 6.0 mmol/L), 110-198 mg/dL (6.1-11.0 mmol/L), 199-251 mg/dL (11.0-13.9 mmol/L) [A], ≥ 252 mg/dL (≥ 14.0	2,471	A: 132 B: 147	2,192	In-hospital	Complications, length of stay

	mmol/L) [B]					
Rueda, 2010	Admission glucose: 60-125 mg/dL (3.3-6.9 mmol/L), 126-179 mg/dL (7.0-9.9 mmol/L) [A], ≥ 180 mg/dL (≥ 10.0 mmol/L) [B]	233	A: 73 B: 32	128	30-day	ICU admission
Schuetz, 2014	Admission glucose: < 108 mg/dL (< 6.0 mmol/L), 108-198 mg/dL (6.0-11.0 mmol/L) [A], > 198 mg/dL (> 11.0 mmol/L) [B]	880	A: 571 B: 93	216	30-day	Severe outcome, ICU admission
Yende, 2010	Admission glucose > 200 mg/dl (> 11.1 mmol/L)	1,795	Unc.	Unc.	1-year	N/A
Iroezindu, 2016	Random blood glucose > 180 mg/dL (> 10.0 mmol/L)	309	59	250	In-hospital	N/A

Supplementary table 3: Details of the seventeen studies associating hyperglycaemia with hospitalized community-acquired pneumonia outcomes. Total numbers are those in whom hyperglycaemia as per the study definition was assessed and may thus be lower than the total number of participants reported in the study. HG = hyperglycaemia. Unc. = unclear. CV = cardiovascular. ICU = intensive care unit. Hyperglycaemia cut-offs = A: ≤ 7.8 mmol/L, B: 7.9-11.0 mmol/L, C: ≥ 11.1 mmol/L, D: not defined.